

Project Proposal

CivicLens: Tender & Police Tracking Web Application

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1. Project Title

CivicLens: Tender & Police Tracking Web Application

2. Type of Project

Type: Web Application

CivicLens is a centralized web-based platform designed to manage public tender activities and police request tracking digitally. The system enables citizens, authorities, contractors, and officers to interact efficiently through an online platform.

3. Objectives

The main objectives of the project are:

- Ensure transparency in public tender management
- Simplify police verification and GD tracking
- Develop a user-friendly online tracking platform
- Improve communication between citizens and government authorities
- Reduce manual paperwork and delays
- Provide real-time notifications and status updates
- Maintain secure and structured data management

4. Problem Statement

Currently, many public service processes are inefficient and difficult to track. Citizens often face delays and lack transparency in government-related activities.

The major problems include:

- Tender information is not easily accessible to the public
- Tender progress and contractor activities are difficult to monitor
- Police verification and complaint handling processes are time-consuming
- Citizens do not receive proper updates regarding their requests or cases
- Manual data handling increases the chance of errors and mismanagement

- Communication gaps exist between users and authorities

CivicLens aims to solve these issues by introducing a digital, centralized, and transparent management system.

5. Deliverables

The expected deliverables of the project are:

- Fully functional web application
- User authentication and verification system
- Tender management dashboard
- Police request submission module
- Officer and authority management system
- Tender assignment and tracking system
- Crime/case assignment and tracking module
- Real-time notification system
- Database schema and ERD design
- Admin panel for system monitoring
- Documentation and project report

6. Methodology

The project will follow the Agile Development Methodology.

Development Process

1. Requirement Analysis
2. System Design and Database Modeling
3. Frontend Development
4. Backend API Development
5. Database Integration

6. Testing and Debugging
7. Deployment and Final Evaluation

Reasons for Using Agile

- Supports iterative development
- Allows continuous improvement
- Makes team collaboration easier
- Helps identify and fix problems quickly
- Flexible for future modifications

7. Project Overview

CivicLens is designed as an integrated public service management system where multiple users can interact through a single platform.

User Features

- Register and log into the system
- Submit police requests and complaints
- Track tender information and updates
- Receive notifications regarding request status
- Access personal dashboard and profile

Authority Features

- Approve or manage tenders
- Monitor contractor activities
- Assign officers to cases or tenders
- Update case progress and status
- Manage public service records

Admin Features

- Manage users and authorities
- Monitor system activities
- Generate audit logs
- Maintain overall system security and integrity

8. Scope

Included Features

- User authentication and verification
- Tender creation and tracking
- Police request and case tracking
- Notification system
- Officer assignment management
- Contractor management
- Database relationship management
- Dashboard and reporting system

Excluded Features

- Physical police operations
- Real-world financial transaction processing
- Mobile application version
- Government-level deployment infrastructure
- AI-based prediction systems

9. Technology Stack

Frontend

- React.js / Next.js
- Streamlit

Backend

- FastAPI or PHP

Database

- MySQL

Authentication

- Firebase Authentication / Supabase

Tools

- Draw.io for ERD and Schema Design
- Git & GitHub for Version Control
- VS Code as Development Environment

10. Risks and Dependencies

Risks

- Server downtime or hosting issues
- Database security vulnerabilities
- API integration problems
- Delays during testing and debugging
- Team coordination challenges

Dependencies

- Stable internet connection
- MySQL database server
- Firebase/Supabase authentication services
- Backend API support
- Browser compatibility

11. Database and ERD Overview

The project database is designed using relational database principles with multiple entities and relationships.

Main Entities

- Users
- User Profiles
- User Verification
- Notifications
- Audit Logs
- Tenders
- Tender Categories
- Tender Documents
- Tender Updates
- Contractors
- Authorities
- Officers
- Police Requests
- Crime Categories

- Crime Assignments
- Case Documents
- Case Updates
- Admins

Relationship Features

- One-to-many relationships
- Foreign key constraints
- Assignment tracking
- Status monitoring
- Secure data integrity

12. Conclusion

CivicLens is designed to improve transparency, accountability, and efficiency in public services by integrating tender management and police tracking into a single web-based platform.

The system provides:

- Structured data management
- Real-time tracking and updates
- Improved communication between citizens and authorities
- Secure and scalable architecture
- User-friendly interaction system

With modular architecture and scalable design, CivicLens can be extended in the future for real-world government and smart-city integration.